

Project on Hamiltonian Simulation - Part II

Theory Exercises

Exercise 1: An Algorithm for Trotterised Hamiltonian Simulation

Exercise 1.1: Algorithm Sketch

Recall the Trotter formula:

$$e^{-iHt} \approx \left(\prod_P e^{-ia_P Pt/r} \right)^r \quad (1)$$

Given the helper function `qc.pauli(P, theta)` implementing $e^{iP\theta}$, consider the following pseudocode:

```
procedure trotter_approx(H, r, t):
    # H is a list of (coefficient, pauli_string) pairs;
    # initialise quantum circuit with n qubits,
    # with n = Pauli string length
    n = len(H[0][1])
    qc = quantum_circuit(n)

    for step in range(r):
        for (a_P, P) in H:
            theta = -a_P * t / r
            qc.pauli(P, theta)

    return qc
```

Explanations:

- The outer range(r) ensures the power of r is applied
- The product over P, or application of all gates, is done in the inner loop using the helper function
- For theta, we need $e^{-ia_P Pt/r}$, so we set $\theta = -a_P t/r$ so that $e^{iP\theta} = e^{iP(-a_P t/r)} = e^{-ia_P Pt/r}$

Exercise 1.2: Why Pauli Decomposition Format?

There is a dimensionality problem when providing the full n qubit Hamiltonian.

Since the Hamiltonian is a matrix of dimensions $2^n \times 2^n$, the number of entries to store is $O(4^n)$. For large n (e.g. $n > 30$), this gets exponentially large and becomes impossible to store!

On the other hand, for a Pauli decomposition of length n , and a Hamiltonian H of $\text{poly}(n)$ terms, we store $O(2 * \text{poly}(n) * n) = O(\text{poly}(n))$ terms. This is much more manageable, as it is polynomial in length (and not exponential)!

The circuit then has $m e^{-ia_P P t/r}$ exponentials constructed with $O(n)$ gates, repeated r times, yielding complexity $O(mrn)$, polynomial in size. This is much better than the exponential complexity of the Hamiltonian derived earlier.

Exercise 2: The Transverse-Field Ising Hamiltonian

Recall the transverse-field Ising Hamiltonian formula:

$$H_{\text{Ising}} = -J \sum_{i=0}^{n-1} Z_i Z_{i+1 \pmod n} - h \sum_i X_i \quad (2)$$

2.1 Interaction Terms

Part (a): proof that $Z_i Z_j$ is diagonal

Lemma: The tensor product of diagonal matrices is diagonal.

Proof: Let $A = \text{diag}(a_1, \dots, a_m)$ and $B = \text{diag}(b_1, \dots, b_n)$ be diagonal matrices.

We write the Kronecker product $A \otimes B$ in block form:

$$A \otimes B = \begin{pmatrix} A_{11}B & A_{12}B & \cdots & A_{1m}B \\ A_{21}B & A_{22}B & \cdots & A_{2m}B \\ \vdots & \vdots & \ddots & \vdots \\ A_{m1}B & A_{m2}B & \cdots & A_{mm}B \end{pmatrix} \quad (3)$$

Since A is diagonal, $A_{ij} = 0$ for $i \neq j$, so all off-diagonal blocks vanish:

$$A \otimes B = \begin{pmatrix} a_1 B & 0 & \cdots & 0 \\ 0 & a_2 B & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_m B \end{pmatrix} \quad (4)$$

Since B is also diagonal, each block $a_i B$ is diagonal. Therefore, $A \otimes B$ is block-diagonal with diagonal blocks, making the entire matrix diagonal. $\square$

Proof of part (a): By the lemma, since I and Z are diagonal, and Z_i, Z_j are tensor products of diagonal matrices, they are both diagonal in the computational basis.

Part (b): Action on computational basis states

We compute how $e^{i(Jt/r)Z_i Z_{i+1}}$ acts on basis states $|a\rangle_i |b\rangle_{i+1}$ for $a, b \in \{0, 1\}$.

First, recall that the Pauli Z matrix acts as:

$$Z |0\rangle = |0\rangle \quad (5)$$

$$Z |1\rangle = -|1\rangle \quad (6)$$

For the product $Z_i Z_{i+1}$:

$$(Z_i Z_{i+1}) |0\rangle_i |0\rangle_{i+1} = Z_i |0\rangle \otimes Z_{i+1} |0\rangle = |0\rangle \otimes |0\rangle = (+1) |00\rangle \quad (7)$$

$$(Z_i Z_{i+1}) |0\rangle_i |1\rangle_{i+1} = Z_i |0\rangle \otimes Z_{i+1} |1\rangle = |0\rangle \otimes (-|1\rangle) = (-1) |01\rangle \quad (8)$$

$$(Z_i Z_{i+1}) |1\rangle_i |0\rangle_{i+1} = Z_i |1\rangle \otimes Z_{i+1} |0\rangle = (-|1\rangle) \otimes |0\rangle = (-1) |10\rangle \quad (9)$$

$$(Z_i Z_{i+1}) |1\rangle_i |1\rangle_{i+1} = Z_i |1\rangle \otimes Z_{i+1} |1\rangle = (-|1\rangle) \otimes (-|1\rangle) = (+1) |11\rangle \quad (10)$$

The eigenvalues are $\lambda_{ab} = (-1)^{a+b}$: $+1$ when $a = b$, and -1 when $a \neq b$.

Applying the definition of the matrix exponential to $e^{i(Jt/r)Z_i Z_{i+1}}$ and using the eigenvalues computed above:

$$e^{i(Jt/r)Z_i Z_{i+1}} |a\rangle_i |b\rangle_{i+1} = \sum_{k=0}^{\infty} \frac{(i(Jt/r))^k (Z_i Z_{i+1})^k}{k!} |a\rangle_i |b\rangle_{i+1} \quad (11)$$

$$= \sum_{k=0}^{\infty} \frac{(i(Jt/r))^k (-1)^{k(a+b)}}{k!} |a\rangle_i |b\rangle_{i+1} \quad (12)$$

$$= e^{i(Jt/r)(-1)^{a+b}} |a\rangle_i |b\rangle_{i+1} \quad (13)$$

By substituting a and b, we get:

$$e^{i(Jt/r)Z_i Z_{i+1}} |0\rangle |0\rangle = e^{i(Jt/r)} |0\rangle |0\rangle \quad (14)$$

$$e^{i(Jt/r)Z_i Z_{i+1}} |0\rangle |1\rangle = e^{-i(Jt/r)} |0\rangle |1\rangle \quad (15)$$

$$e^{i(Jt/r)Z_i Z_{i+1}} |1\rangle |0\rangle = e^{-i(Jt/r)} |1\rangle |0\rangle \quad (16)$$

$$e^{i(Jt/r)Z_i Z_{i+1}} |1\rangle |1\rangle = e^{i(Jt/r)} |1\rangle |1\rangle \quad (17)$$

2.2 Transverse Field Terms

Part (a): Diagonalizing basis

Recall the Pauli X matrix has eigenvectors:

$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle), \quad \text{eigenvalue } +1 \quad (18)$$

$$|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle), \quad \text{eigenvalue } -1 \quad (19)$$

Applying the transverse field term $e^{i(ht/r)X_i}$ to each state:

$$e^{i(ht/r)X_i} |+\rangle = \sum_{k=0}^{\infty} \frac{(i(ht/r)X_i)^k}{k!} |+\rangle \quad (20)$$

$$= \sum_{k=0}^{\infty} \frac{(i(ht/r))^k X_i^k}{k!} |+\rangle \quad (21)$$

$$= \sum_{k=0}^{\infty} \frac{(i(ht/r))^k \cdot 1^k}{k!} |+\rangle \quad (\text{since } X_i |+\rangle = |+\rangle) \quad (22)$$

$$= \left(\sum_{k=0}^{\infty} \frac{(i(ht/r))^k}{k!} \right) |+\rangle \quad (23)$$

$$= e^{i(ht/r)} |+\rangle \quad (24)$$

Similarly:

$$e^{i(ht/r)X_i} |-\rangle = \sum_{k=0}^{\infty} \frac{(i(ht/r))^k X_i^k}{k!} |-\rangle \quad (25)$$

$$= \sum_{k=0}^{\infty} \frac{(i(ht/r))^k \cdot (-1)^k}{k!} |-\rangle \quad (\text{since } X_i |-\rangle = -|-\rangle) \quad (26)$$

$$= e^{-i(ht/r)} |-\rangle \quad (27)$$

Therefore $e^{i(ht/r)X_i}$ is diagonal in the X -basis $\{|+\rangle, |-\rangle\}$, with eigenvalues $e^{\pm i(ht/r)}$.

Part (b): Relation to rotation gate and bit flip

Recall the rotation gate's definition:

$$R_X(\theta) = e^{-i\theta X/2} \quad (28)$$

Compare with $e^{i(ht/r)X_i}$:

$$e^{i(ht/r)X_i} = e^{-i\theta X_i/2} \quad (29)$$

Matching exponents:

$$i(ht/r) = -i\theta/2 \quad \Rightarrow \quad \theta = -\frac{2ht}{r} \quad (30)$$

Therefore:

$$e^{i(ht/r)X_i} = R_{X_i} \left(-\frac{2ht}{r} \right) \quad (31)$$

When the rotation gate acts as a bit flip

For $e^{i(ht/r)X_i}$ to act proportionally to X_i , we need some scalar c such that:

$$e^{i(ht/r)X_i} |+\rangle = c \cdot X_i |+\rangle = c |+\rangle \quad (32)$$

$$e^{i(ht/r)X_i} |-\rangle = c \cdot X_i |-\rangle = -c |-\rangle \quad (33)$$

From part (a), this requires:

$$e^{i(ht/r)} = c \quad (34)$$

$$e^{-i(ht/r)} = -c \quad (35)$$

Substituting the first equation into the second:

$$e^{-i(ht/r)} = -e^{i(ht/r)} \iff 2 \cos(ht/r) = 0 \iff ht/r = \frac{\pi}{2} + k\pi, \quad k \in \mathbb{Z} \quad (36)$$

Therefore:

$$t = \frac{\pi r(2k+1)}{2h}, \quad k \in \mathbb{Z}_+ \quad (37)$$

Exercise 3: Hamiltonian Simulation of the Transverse-Field Ising Hamiltonian

Exercise 3.1: Circuit Implementation

See the jupyter notebook.

Exercise 3.2: Comparison Method

In order to compare two circuits for "closeness" (i.e. whether they are approximately the same), we can use distance defined through the operator norm:

$$D(U, V) = \|U - V\|_{\text{op}} \quad (38)$$

Where the operator norm $\|M\|_{\text{op}}$ is defined as $\max_{\|\psi\|=1} \|A|\psi\rangle\|_2$.

Indeed, if $U = V$, we have $D(U, V) = 0$ as required. Moreover, it is the same distance used in part 1. It is classical (i.e. extractable through `Operator(qc).data` in Qiskit), but sadly inefficient, as the full unitary scales exponentially with n .

Exercise 3.3: Comparison Results vs. Theoretical Bound

The theoretical bound for $n = 4$, $J = h = 1$, $t = 5$ and $m = 2n = 8$ is:

$$O\left(\frac{t^2}{r} \cdot m^2 \cdot \max_P |a_P|^2\right) = O\left(\frac{t^2}{r} \cdot 16 \cdot 1\right) = O\left(\frac{16t^2}{r}\right) \quad (39)$$

Expected error trend: since the theoretical bound scales with $1/r$, the error should decrease as $r \rightarrow \infty$.

Observed error trend: plotting the error defined in 3.2 reveals a decreasing trend, which aligns with the $1/r$ scaling.

Exercise 3.4(d): Interpretation of $p_0(t)$ Plots

The plots of $p_0(t)$ vary depending on h :

- $h = 0.2$: we have $h < J$, meaning the X_i terms don't affect the system as much; $p_0(t) \approx 1$ with a small Trotter error given that $\max_P |a_P|^2 = \max(J^2, h^2) = J^2 = 1$. Most r values align well with the exact solution, with only $r = 1, 2$ showing noticeable deviation.
- $h = 1$: we have $h = J$, which translates to considerable oscillations; there is strong divergence with small r (i.e. $r = 1, 2$), and overall alignment with the exact solution is not reached for most r values (except $r = 50$, which tracks the exact solution very well).
- $h = 2$: strong divergence at small r values, with significant and frequent oscillations. Likely caused by the term $\max(J^2, h^2) = h^2 = 4$ in the error bound. Similarly to $h = 1$, $r = 50$ tracks the exact solution closely.

Exercise 3.4(e): Is Fixed r Sensible?

No; as t increases with fixed r , the error bound $O(t^2/r \cdot m^2 \max_P |a_P|^2)$ grows boundlessly, scaling with t^2 , resulting in a less accurate approximation. To counter this effect, we should choose $r \propto t^2$.

Exercise 3.4(f): Time-Dependent $r(t)$

Since we require $r(t) \propto t^2$ with $r(T) = r(5) = 30$, we set:

$$r(t) = \left\lceil \frac{6}{5}t^2 \right\rceil \tag{40}$$

This way, the ceiling ensures getting an integer result, and $r(5) = \lceil \frac{6}{5} \cdot 25 \rceil = 30$. To avoid divisions by 0, we clip to $\max(1, \cdot)$ in the implementation.

Simulation results: plotting the simulation with time-dependent $r(t)$ shows more uniform alignment with the exact solution overall, with certain anomalies close to $r(t) = 1$. These results are noticeably better than those with fixed r , which shows using a time-dependent $r(t)$ is the better choice.